

## Medical Image Processing M2 Data Science & AI 2024-2025

[www.sop.inria.fr/teams/asclepios/cours/MS\\_DS/](http://www.sop.inria.fr/teams/asclepios/cours/MS_DS/)

X. Pennec

### Diffeomorphic deformations and computational anatomy



26/02/2025

Epione team  
2004, route des Lucioles B.P. 93  
06902 Sophia Antipolis Cedex  
<http://www.sop.inria.fr/epione>

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## Medical Image Analysis – M2 DSAI 2024-2025

Thursday afternoon

Course notes : [http://www.sop.inria.fr/teams/asclepios/cours/MS\\_DS/](http://www.sop.inria.fr/teams/asclepios/cours/MS_DS/)

- Thur. Jan. 09, 14:00-17:00 [Xavier Pennec] Introduction to Medical Image Acquisition
- Thur. Jan. 16 13:30-16:30 [Xavier Pennec] Medical Image Registration
- Thur. Jan. 23 13:30-16:30 [Hervé Delingette] Image Filtering
- Thur. Jan. 30 13:30-16:30 [Xavier Pennec] Statistics on Riemannian manifolds and Lie groups
- Thur. Feb. 6 13:30-16:30 [Hervé Delingette] Principles and Metrics for Image segmentation
- Thur. Feb. 13 13:30-16:30 [Hervé Delingette] Image segmentation based on clustering
- Thur. Feb. 27 13:30-16:30 [Xavier Pennec] Diffeomorphic Registration and Computational Anatomy
- Thur. March 6, 13:30-16:30 [Hervé Delingette] Connexity and Shape Constrained Image segmentation
- Thur. March 13, 13:30-15:00, [XP & HD] EXAM

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2

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## Statistical Computing on Manifolds for Computational Anatomy

### Metric and Affine Geometric Settings for Lie Groups

- Deformable image registration
- Riemannian frameworks on Lie groups
- Lie groups as affine connection spaces

### Modeling longitudinal deformations in AD

- The SVF framework for diffeomorphisms
- Parallel transport of deformation trajectories

### Perspectives for statistics on deformation

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3

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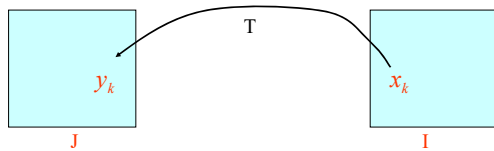
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## Goals of Registration

### A dual problem

- Find the point  $y$  of image  $J$  which is corresponding (homologous) to each points  $x$  of image  $I$ .
- Determine the best transformation  $T$  that superimposes homologous points

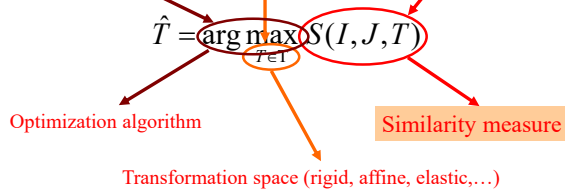


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4

## Mathematical Formulation of registration (Brown, 1992)

■ **Registration:** Given two datasets (images)  $I$  and  $J$ , find the geometric transformation  $T$  that « best » aligns the physically homologous points (voxels)



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5

## The deformable registration landscape in 1995

**Transformation encoded by a displacement field:**  $T(x) = x + u(x)$

**Optical flow**  $F(x, u) = -(I(x) - J(x + u)) \nabla J(x + u)$

- Horn and Schunck, Artif. Intell. 17, 1981;
- Aggarwal and Nandhakumar, Proc. IEEE 76, 1988;
- Barron et al., 1994.

$$\frac{\partial u}{\partial t} \propto F(x, u)$$

**Linear elastic deformation**  $\mu \nabla^2 u + (\mu + \lambda) \nabla (\text{div}(u)) = F$

- Broit, PhD 1981.
- Bajcsy and Kovacic CVGIP 46, 1989
- Gee, Reivich, Bajcsy, J. Comp. Assis. Tom. 17, 1993.

**Fluid (images & surface)**  $\mu \nabla^2 v + (\mu + \lambda) \nabla (\text{div}(v)) = F$

- Christensen, Rabbitt, Miller, Phys. Med. Biol. 39, 1994.
- Christensen, Rabbitt, Miller, IEEE TIP. 5(10), 1996.
- Thompson and Toga, IEEE TMI 15(4), 1996.

$$\frac{\partial u}{\partial t} = v - (\nabla u) v$$

Differential equations were costly to solve: > 1 day on mass-parallel machine

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6

***Demons' algorithm (MRCAS 95, CVPR96, Media98)***

Medical Image Analysis (1996) volume 2, number 3, pp 243-269  
© Oxford University Press

### Image matching as a diffusion process: an analogy with Maxwell's demons

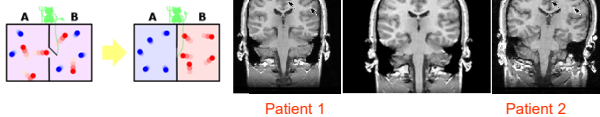
J. P. Thirion\*

INRIA, Equipe Épidémié, 2004 Route des Lucioles BP933, 06902 Sophia-Antipolis, France

**Abstract**

In this paper, we present the concept of offloading models to perform image-to-image mapping through two images to match. The main idea is to consider the object boundaries in one image as point correspondences between the two images. In the other strategy, considered in a following grid-based approach, we use the boundaries of the objects in the second image as point correspondences. We illustrate this concept by an analogy with Muesli's dilemma. We show that this concept relates to more traditional ones, based on estimation, with an intermediate step being applied to the images. We use the concept of offloading models to derive three different non-local matching algorithms. As a by-product, we use the analogy between the point-to-point, the early point-to-line, and a last one repeating on already registered images. Finally, we present results with synthesized deformations and real medical images, with applications to heart motion tracking and three-dimensional intra-organ motion tracking.

Received October 22, 1990; revised August 8, 1991; March 16, 1992; accepted April 13, 1992.



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7

***Demons' algorithm (MRCAS 95, CVPR96, Media98)***

- $T_0 = \text{Identity}$

- Update field

$$U_{n+1} = \frac{I - J \circ T_n}{\|\nabla I\|^2 + (I - J \circ T_n)^2} \nabla I$$

- Regularization by Gaussian filtering

Elastic

$$\hat{T}_{n+1} = T_n \circ U_{n+1}$$

$$T_{n+1} = G_\sigma * \hat{T}_{n+1}$$

Fluid

$$\tilde{U}_{n+1} = G_\sigma * U_{n+1}$$

$$T_{n+1} = T_n \circ \tilde{U}_{n+1}$$

Why does that work? Convergence? Change the similarity metric?

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8

### *Interpretation of demons*

$$E(C,T) = SSD(I,J,C) + \sigma \|C - T\|^2 + \sigma \lambda. \text{Reg}(T)$$

- $S_{SD}$  : measures the similarity of intensities
  - $R_{eg}$  : regularization energy (quadratic)
  - $\lambda, \sigma$  : smoothing and noise parameters
  - $C$  : correspondences between points (vectors field)
  - $T$  : transformation (regularized vector field)
- Introduce correspondences (matches) as an auxiliary variable to decouple into a local non-convex

P. Cachier E. Bardinet, D. Dormont, XP and N. Ayache: *Iconic Feature Based Nonrigid Registration: the PASHA Algorithm*, Comp. Vision and Image Understanding (CVIU), Special Issue on Non Rigid Registration, 89 (2-3), 272-298, 2003.

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9

## PASHA Algorithm (2/2)

$$E(C, T) = SSD(I, J, C) + \sigma \|C - T\|^2 + \sigma \lambda \text{Reg}(T)$$

### Alternated minimization

- Minimization with respect to  $C$  :
  - Find matches between points by optimizing  $E_S$  + in the neighborhood of  $T$
  - Gradient descent (1<sup>st</sup>, 2<sup>nd</sup> order, e.g. Gauss-Newton)
- Minimization with respect to  $T$  :
  - Find a smooth transformation that approximates  $C$
  - Quadratic energy  $\Rightarrow$  convolution
- **Interest:** fast computation

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10

## Newton optimization of the correspondence energy

$$E(C) = \int (I(x) - J(C(x)))^2 dx + \frac{\sigma_I^2}{\sigma_x^2} \int \|C(x) - T(x)\|^2 dx$$

### Exact solution of the quadratic approximation of the SSD

- Solve  $\left[ (\nabla J \circ T)(\nabla J \circ T)^T + \frac{\sigma_I^2}{\sigma_x^2} Id \right] u = (J \circ T - I)(\nabla J \circ T)$
- By inversion lemma:  $u = \frac{(J \circ T - I)(\nabla J \circ T)}{\|\nabla J \circ T\|^2 + \sigma_I^2 / \sigma_x^2}$
- Local estimation of intensity variance:  $\sigma_I^2 = (J \circ T - I)^2$
- Assuming isotropic voxel size:  $\sigma_x^2 \approx 1$

$$u = \frac{I - J \circ T}{\|\nabla I\|^2 + (I - J \circ T)^2} \nabla I$$

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11

## Efficient Regularization

**Quadratic regularizer**  $\text{Reg}(T) = \int \sum_{k=1}^{\infty} \frac{\sum_{i_1 \dots i_k} \partial_{i_1} \dots \partial_{i_k} (T - Id)}{\sigma_d^{2k} k!} \|^2$

**Euler Lagrange optimization of**  $E(T) = \int \|C - T\|^2 + \text{Reg}(T)$

$$C - T + \sum_{k=1}^{\infty} \frac{(-1)^k \Delta^k (T - Id)}{\sigma_d^{2k} k!} = 0$$

**Solution: Gaussian smoothing**  $T_{\text{opt}} = G_{\sigma} * C$  with  $\sigma = 1 / \sigma_d$

Pennece, Cachier, Ayache. Understanding the "Demon's Algorithm": 3D Non-Rigid registration by Gradient Descent. MICCAI 1999.

### Extension to a family of quadratic filters

$$G_{\sigma, \kappa}(\mathbf{u}) = \frac{1}{(\sigma \sqrt{2\pi})^3 (1 + \kappa)} \left( Id + \frac{\kappa}{\sigma^2} \mathbf{u} \mathbf{u}^T \right) \exp \left( -\frac{\mathbf{u}^T \mathbf{u}}{2\sigma^2} \right)$$

P. Cachier and N. Ayache. Isotropic energies, filters and splines for vectorial regularization. J. of Math. Imaging and Vision, 20(3):251-265, May 2004.

26/02/2025

12

### Inter-subject registration

#### Affine transformation

MR T1 Images  
256x256x120 voxels  
Atlas to patient registration  
for radiotherapy planning

Correct size and position but high remaining variability in cortex and deep structures

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13

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### Inter-subject registration

#### Fluid regularization

Very good image correspondence

But anatomically meaningless deformation  
Jacobian [1/50;50]

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14

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### Inter-subject registration

#### Adaptive non-stationary visco-elastic regularization

Registration in 5 min on 15 PCs

Anatomically more meaningful deformation  
Jacobian [1/5;5]

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15

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## Statistical Computing on Manifolds for Computational Anatomy

### Metric and Affine Geometric Settings for Lie Groups

- Deformable image registration
- Riemannian frameworks on Lie groups
- Lie groups as affine connection spaces

### Modeling longitudinal deformations in AD

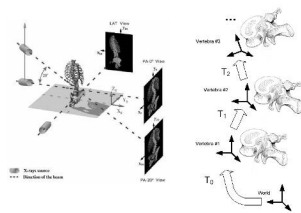
- The SVF framework for diffeomorphisms
- Parallel transport of deformation trajectories

### Perspectives for statistics on deformation

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1

## Statistical Analysis of the Scoliotic Spine



### Data

- 307 Scoliotic patients from the Montreal's St-Justine Hosp
- 3D Geometry from multi-planar X-rays
- Articulated model: 17 relative pose of successive vertebrae

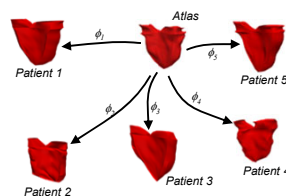
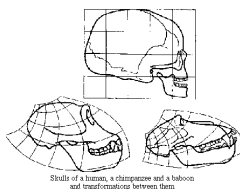
### Statistics

- Main translation variability is axial (growth?)
- Main rot. var. around anterior-posterior axis
- 4 first variation modes related to King's classes

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[ J. Boisvert et al. ISBI'06, AMD'06 and IEEE TMI 27(4), 2008 ]

## Morphometry through Deformations



### Measure of deformation [D'Arcy Thompson 1917, Grenander & Miller]

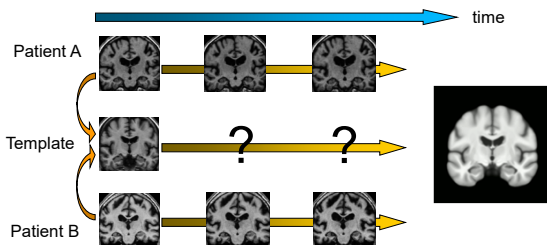
- Observation = "random" deformation of a reference template
- Deterministic template = anatomical invariants [Atlas ~ mean]
- Random deformations = geometrical variability [Covariance matrix]

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18

## Longitudinal deformation analysis

### Deformation trajectories in different reference spaces



Mean longitudinal deformation across subjects?  
Convenient mathematical settings for transformations?

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19

## Statistics on deformations

### Statistics on displacement field/transformation parameters

- Splines [Rueckert et al., TMI, 03],
- PCA of Statistical shape models
- Simple vector statistics, but inconsistency with group properties

### The Riemannian approach (LDDMM)

- Right-invariant metric on diffeos [Joshi, Miller, Trounev, Younes...]
- Parameterize diffeomorphisms by time-varying velocity fields
- Good mathematical bases for statistics on non-linear spaces

### No bi-invariant metric in general

- Left/right Fréchet mean incompatible with group structure
- The inverse of the mean is not the mean of the inverse
- Examples with simple 2D rigid transformations

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20

## General Non-Compact and Non-Commutative case

### No Bi-invariant Mean for 2D Rigid Body Transformations

- Metric at Identity:  $\text{dist}(Id, (\theta; t_1; t_2))^2 = \theta^2 + t_1^2 + t_2^2$
- $T_1 = \left(\frac{\pi}{4}; -\frac{\sqrt{2}}{2}; \frac{\sqrt{2}}{2}\right)$   $T_2 = (0; \sqrt{2}; 0)$   $T_3 = \left(-\frac{\pi}{4}; -\frac{\sqrt{2}}{2}; -\frac{\sqrt{2}}{2}\right)$
- Left-invariant Fréchet mean:  $(0; 0; 0)$
- Right-invariant Fréchet mean:  $\left(0; \frac{\sqrt{2}}{3}; 0\right) \approx (0; 0.4714; 0)$

### Questions for this talk:

- Can we design a mean compatible with the group operations?
- Is there a more convenient structure for statistics on Lie groups?

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21

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2

## Basics of Lie groups

### Flow of a left invariant vector field $\tilde{X} = DL.x$ starting from $e$

- $\gamma_x(t)$  exists for all time
- One parameter subgroup:  $\gamma_x(s+t) = \gamma_x(s) \cdot \gamma_x(t)$

### Lie group exponential (ATTN: different from Riemannian Exp)

- Definition:  $x \in \mathfrak{g} \rightarrow \text{Exp}(x) = \gamma_x(1) \in G$
  - Diffeomorphism from a neighborhood of 0 in  $\mathfrak{g}$  to a neighborhood of  $e$  in  $G$  (not true in general for inf. dim)
  - Baker-Campbell Hausdorff (BCH) formula
- $$\text{BCH}(x, y) = \text{Log}(\text{Exp}(x) \cdot \text{Exp}(y)) = x + y + \frac{1}{2}[x, y] + \dots$$

### 3 curves at each point parameterized by the same tangent vector

- Left / Right-invariant geodesics, one-parameter subgroups

### Question: Can one-parameter subgroups be geodesics?

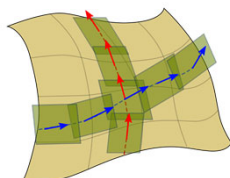
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23

## Affine connection spaces

### Affine Connection (infinitesimal parallel transport)

- Acceleration = derivative of the tangent vector along a curve
- Projection of a tangent space on a neighboring tangent space



### Geodesics = straight lines

- Null acceleration:  $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$
- 2<sup>nd</sup> order differential equation:  
Normal coordinate system
- Local exp and log maps (Strong form of Whitehead theorem:  
In an affine connection space, each point has a normal convex neighborhood  
(unique geodesic between any two points included in the NCN))

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24



## Canonical Connections on Lie Groups

### A unique Cartan-Schouten connection

- Symmetric (no torsion) and bi-invariant
- For which geodesics through Id are one-parameter subgroups (group exponential)
  - Matrices:  $M(t) = A \cdot \exp(t \cdot V)$
  - Diffeos: **left/right translations of Stationary Velocity Fields (SVFs)**

### Levi-Civita connection of a bi-invariant metric (if it exists)

- Continues to exist in the absence of such a metric (e.g. for rigid or affine transformations)

### Symmetric space with central symmetry $S_\psi(\phi) = \psi\phi^{-1}\psi$

- Matrix geodesic symmetry:  $S_A(M(t)) = A \exp(-tV)A^{-1}A = M(-t)$

[Lorenzi, Pennec. Geodesics, Parallel Transport & One-parameter Subgroups for Diffeomorphic Image Registration. Int. J. of Computer Vision, 105(2):111-127, 2013.]

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25

## Mean value on an affine connection space

### Fréchet mean cannot be defined (no distance) but:

$$E[\mathbf{x}] = \underset{y \in M}{\operatorname{argmin}} \left( E[\operatorname{dist}(y, \mathbf{x})^2] \right) \Rightarrow E[\overline{\mathbf{x}\mathbf{x}}] = \int_M \overline{\mathbf{x}\mathbf{x}} \cdot p_{\mathbf{x}}(z) \cdot dM(z) = 0 \quad [P(C) = 0]$$

### Exponential barycenters

- [Emery & Mokobodzki 91, Corcuera & Kendall 99]
- $\int \operatorname{Log}_x(y) \mu(dy) = 0 \quad \text{or} \quad \sum_i \operatorname{Log}_x(y_i) = 0$
- Existence? Uniqueness?
- OK for convex affine manifolds with semi-local convex geometry
- [Arnaudon & Li, Ann. Prob. 33-4, 2005]
  - Use a separating function (convex function separating points) instead of a distance
- Algorithm to compute the mean: fixed point iteration (stability?)

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26

## Bi-invariant Mean on Lie Groups

### Exponential barycenter of the symmetric Cartan connection

- Locus of points where  $\sum \operatorname{Log}(m^{-1} \cdot g_i) = 0$  (whenever defined)
- Iterative algorithm:  $m_{t+1} = m_t \circ \operatorname{Exp} \left( \frac{1}{n} \sum \operatorname{Log}(m_t^{-1} \cdot g_i) \right)$
- First step corresponds to the Log-Euclidean mean
- Corresponds to the first definition of bi-invariant mean of [V. Arsigny, X. Pennec, and N. Ayache. Research Report RR-5885, INRIA, April 2006.]

### Mean is stable by left / right composition and inversion

- If  $m$  is a mean of  $\{g_i\}$  and  $h$  is any group element, then
  - $h \circ m$  is a mean of  $\{h \circ g_i\}$ ,
  - $m \circ h$  is a mean of the points  $\{g_i \circ h\}$
  - and  $m^{(-1)}$  is a mean of  $\{g_i^{(-1)}\}$

[Pennec & Arsigny, Ch.7 p.123-166, Matrix Information Geometry, Springer, 2012]

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27

### Example mean of 2D rigid-body transformation

$$T_1 = \left( \frac{\pi}{4}, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right) \quad T_2 = (0; \sqrt{2}; 0) \quad T_3 = \left( -\frac{\pi}{4}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right)$$

- ▢ Metric at Identity:  $\text{dist}(Id, (\theta; t_1; t_2))^2 = \theta^2 + t_1^2 + t_2^2$
- ▢ Left-invariant Fréchet mean:  $(0; 0; 0)$
- ▢ Log-Euclidean mean:  $\left( 0; \frac{\sqrt{2}-\pi/4}{3}; 0 \right) \simeq (0; 0.2096; 0)$
- ▢ Bi-invariant mean:  $\left( 0; \frac{\sqrt{2}-\pi/4}{1+\pi/4(\sqrt{2}+1)}; 0 \right) \simeq (0; 0.2171; 0)$
- ▢ Right-invariant Fréchet mean:  $\left( 0; \frac{\sqrt{2}}{3}; 0 \right) \simeq (0; 0.4714; 0)$

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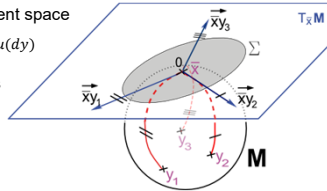
29

### Generalization of the Statistical Framework

#### Covariance matrix & higher order moments

- ▢ Defined as tensors in tangent space
- $$\Sigma = \int \text{Log}_x(y) \otimes \text{Log}_x(y) \mu(dy)$$

- ▢ Matrix expression changes according to the basis



#### Other statistical tools

- ▢ Mahalanobis distance well defined and bi-invariant
- $$\mu_{(m,\Sigma)}(g) = \int [\text{Log}_m(g)]^i \Sigma_{ij}^{-1} [\text{Log}_m(g)]^j \mu(dy)$$
- ▢ Tangent Principal Component Analysis (tPCA)
  - ▢ Principal Geodesic Analysis (PGA), provided a data likelihood
  - ▢ Independent Component Analysis (ICA)

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30

### <http://geomstats.ai> : a python library to implement generic algorithms on many Riemannian manifolds

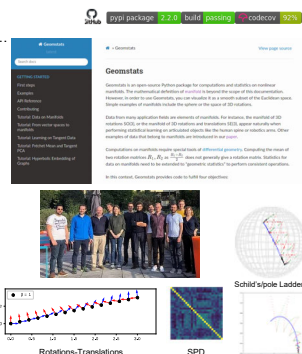
- ▢ Mean, PCA, clustering, parallel transport...

- ▢ 15 manifolds / Lie groups already implemented (SPD, H(n), SE(n), etc)
- ▢ Generic manifolds with geodesics by integration / optimization

- ▢ scikit-learn API (hide geometry, compatible with GPU & learning tools).

- ▢ 10 introductory tutorials
- ▢ ~35000 lines of code
- ▢ ~30 academic developers
- ▢ 7 hackathons organized in 2020-2022
- ▢ Last one: 17-21 October 2022 IHP, Paris

[ Miolane et al, JMLR 2020, Scipy 2020  
Guigui et al, FNT in Mach. Learning 2023 ]



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## Cartan Connections vs Riemannian

### What is similar

- Standard differentiable geometric structure [curved space without torsion]
- Normal coordinate system with  $\text{Exp}_x$  et  $\text{Log}_x$  [finite dimension]

### Limitations of the affine framework

- No metric (but no choice of metric to justify)
- The exponential does always not cover the full group
  - Pathological examples close to identity in finite dimension
  - In practice, similar limitations for the discrete Riemannian framework
- Global existence and uniqueness of bi-invariant mean?  
Use a bi-invariant pseudo-Riemannian metric? [Miolane MaxEnt 2014]

### What we gain

- A globally invariant (composition & inversion) symmetric space structure
- Simple geodesics, efficient computations (stationarity, group exponential)
- The simplest linearization of transformations for statistics?

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32

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### Perspectives for statistics on deformation

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3

## Riemannian Metrics on diffeomorphisms

### Space of deformations

- Transformation  $y = \phi(x)$
- Curves in transformation spaces:  $\phi(x, t)$
- Tangent vector = speed vector field  $v_t(x) = \frac{d\phi(x, t)}{dt}$

### Right invariant metric

$$\|v_t\|_{\phi_t} = \|v_t \circ \phi_t^{-1}\|_{Id}$$

- Eulerian scheme
- Sobolev Norm  $H_k$  or  $H_\infty$  (RKHS) in LDDMM  $\rightarrow$  diffeomorphisms [Miller, Trounev, Younes, Holm, Dupuis, Beg... 1998 – 2009]

### Geodesics determined by optimization of a time-varying vector field

- Distance  $d^2(\phi_0, \phi_1) = \arg \min_{v_t} \left( \int_0^1 \|v_t\|_{\phi_t}^2 dt \right)$
- Geodesics characterized by initial velocity / momentum
- Optimization for images is quite tricky (and lengthy)

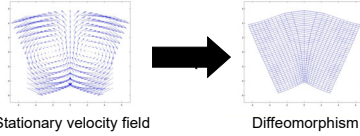
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34

## The SVF framework for Diffeomorphisms

**Idea:** [Arsigny MICCAI 2006, Bossa MICCAI 2007, Ashburner Neuroimage 2007]

- Exponential of a smooth vector field is a diffeomorphism
- Parameterize deformation by ~~time-varying~~ Stationary Velocity Fields



Stationary velocity field

Diffeomorphism

### Direct generalization of numerical matrix algorithms

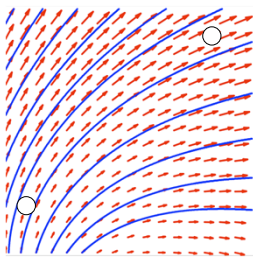
- Computing the deformation: **Scaling and squaring** [Arsigny MICCAI 2006]  
recursive use of  $\exp(v) = \exp(v/2) \circ \exp(v/2)$
- Updating the deformation parameters: **BCH formula** [Bossa MICCAI 2007]  
 $\exp(v) \circ \exp(u) = \exp(v + \epsilon u + [v, \epsilon u]/2 + [v, [v, \epsilon u]]/12 + \dots)$
- Lie bracket  $[v, u](p) = \text{Jac}(v)(p) \cdot u(p) - \text{Jac}(u)(p) \cdot v(p)$

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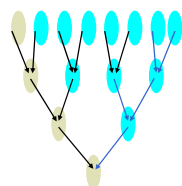
35

## Computing the exponential

$$\exp(u) = \exp(u/N)^N$$



$$\begin{aligned} \frac{\partial x}{\partial t} &= v(x) \\ x(0) &= x_0 \\ x(1) &= \int_0^1 v(x(t)) dt \\ &\triangleq \exp(v) \end{aligned}$$



$$\begin{aligned} \exp(v/8) &\approx \text{Id} + v/8 \\ \exp(v/4) &= \exp(v/8)^2 \\ \exp(v/2) &= \exp(v/4)^2 \\ \exp(v) &= \exp(v/2)^2 \end{aligned}$$

[V. Arsigny, O. Commowick, X. Pennec, N. Ayache. A Log-Euclidean Framework for Statistics on Diffeomorphisms. In Proc. of MICCAI'06, LNCS 4190, pages 924-931, 2-4 October 2006.]

26/02/2025

36

## Symmetric log-demons [Vercauteren MICCAI 08]

**Idea:** [Arsigny MICCAI 2006, Bossa MICCAI 2007, Ashburner Neuroimage 2007]

- Parameterize the deformation by SVFs
- Time varying (LDDMM) replaced by stationary vector fields
- Efficient scaling and squaring methods to integrate autonomous ODEs

### Log-demons with SVFs

$$\mathcal{E}(v, v_c) = \underbrace{\frac{1}{\sigma_f^2} \|F - M \circ \exp(v_c)\|_{L_2}^2}_{\text{Similarity}} + \underbrace{\frac{1}{\sigma_v^2} \|\log(\exp(-v) \circ \exp(v_c))\|_{L_2}^2}_{\text{Coupling}} + \underbrace{\mathcal{R}(v)}_{\text{Regularisation}}$$

Measures how much the two images differ      Couples the correspondences with the smooth deformation      Ensures deformation smoothness

- Efficient optimization with BCH formula
- Inverse consistent with symmetric forces
- Open-source ITK implementation
  - Very fast
  - <http://hdl.handle.net/10380/3060>

[T Vercauteren, et al., *Symmetric Log-Domain Diffeomorphic Registration: A Demons-based Approach*, MICCAI 2008]

26/02/2025

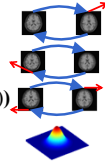
37

### Symmetric Log-Domain Demons

Use easy inverse:  $T^{-1} = \exp(-v)$

#### Iteration

- Given images  $I_0, I_1$  and current transformation  $T = \exp(v)$
- Forward demons forces  $u^{\text{forw}}$
- Backward demons forces  $u^{\text{back}}$
- Update  $v_c \leftarrow \frac{1}{2} (\text{BCH}(v, u^{\text{forw}}) - \text{BCH}(-v, u^{\text{back}}))$
- Regularize (Gaussian):  $v \leftarrow K_{\text{diff}} * v_c$



#### Open-source ITK implementation

- Very fast
  - <http://hdl.handle.net/10380/3060>
- [ T Vercauteren, et al., Symmetric Log-Domain Diffeomorphic Registration: A Demons-based Approach, MICCAI 2008 ]

26/02/2025

38

### Statistical Computing on Manifolds for Computational Anatomy

#### Metric and Affine Geometric Settings for Lie Groups

- Deformable image registration
- Riemannian frameworks on Lie groups
- Lie groups as affine connection spaces

#### Modeling longitudinal deformations in AD

- The SVF framework for diffeomorphisms
- Parallel transport of deformation trajectories

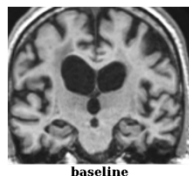
#### Perspectives for statistics on deformation

26/02/2025

3

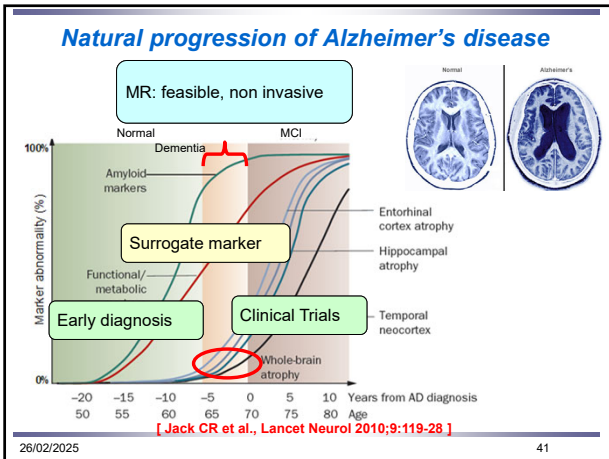
### Alzheimer's Disease

- Most common form of dementia
- 18 Million people worldwide
- Prevalence in advanced countries
  - 65-70: 2%
  - 70-80: 4%
  - 80 - : 20%
- If onset was delayed by 5 years, number of cases worldwide would be halved



26/02/2025

40




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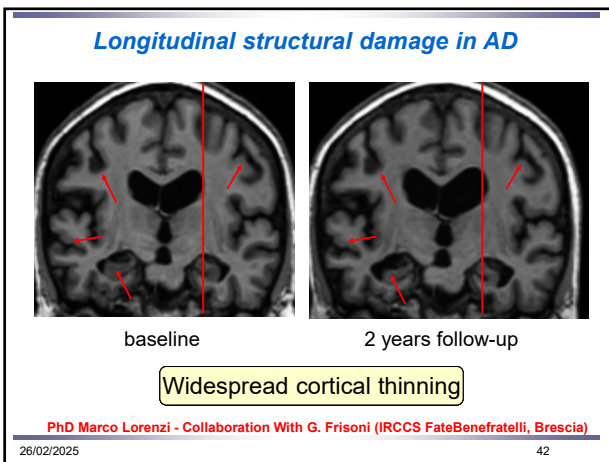
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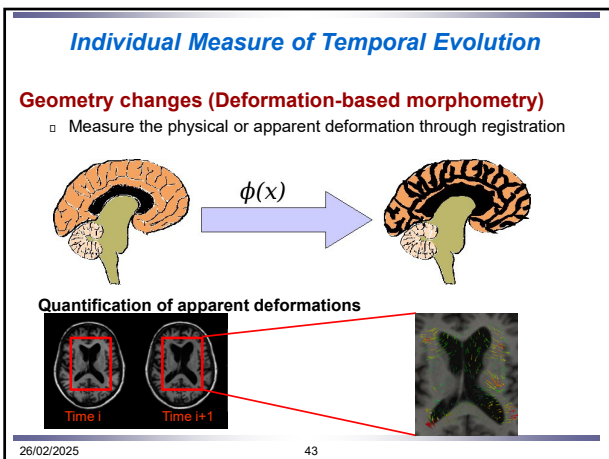
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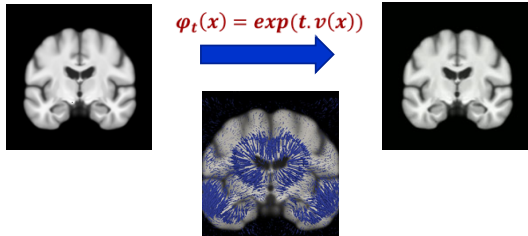
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## Measuring Temporal Evolution with deformations: Deformation-based morphometry

Fast registration with deformation parameterized by SVF



<https://team.inria.fr/asclepios/software/lcclogdemos/>

[ Lorenzi, Ayache, Frisoni, Pennec, Neuroimage 81, 1 (2013) 470-483 ]

26/02/2025

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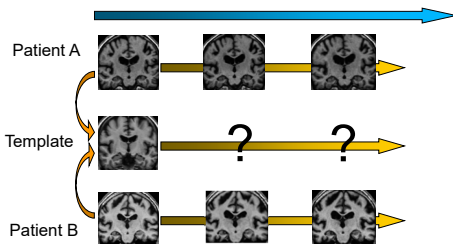
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## Longitudinal deformation analysis in AD

- From patient specific evolution to population trend (parallel transport of deformation trajectories)
- Inter-subject and longitudinal deformations are of different nature and might require different deformation spaces/metrics



PhD Marco Lorenzi - Collaboration With G. Frisoni (IRCCS FateBenefratelli, Brescia)

26/02/2025

45

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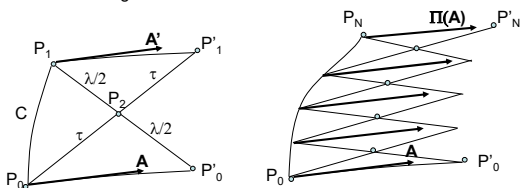
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## From gravitation to computational anatomy: Parallel transport along arbitrary curves

Infinitesimal parallel transport = connection  $\nabla_t(x) : TM \rightarrow TM$

A numerical scheme to integrate symmetric connections:  
Schild's Ladder [Eiher et al, 1972]

- Build geodesic parallelogramoid
- Iterate along the curve



26/02/2025

46

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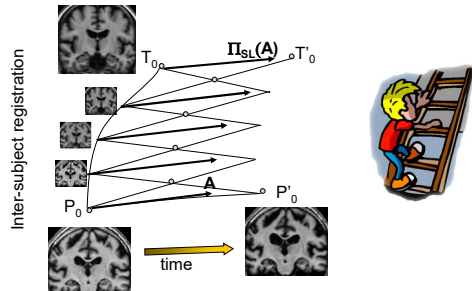
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## Schild's Ladder Intuitive application to images



[Lorenzi, Ayache, Pennec: Schild's Ladder for the parallel transport of deformations in time series of images, IPMI 2011 ]

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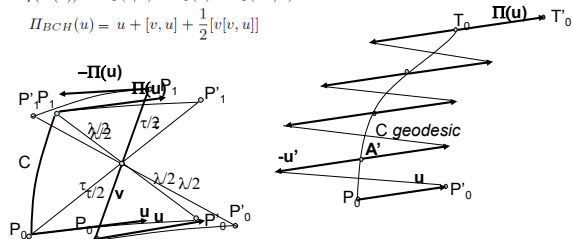
47

## Parallel transport along geodesics

### Simpler scheme along geodesics: Pole Ladder

$$\text{Exp}(\Pi(u)) = \text{Exp}(v/2) \circ \text{Exp}(u) \circ \text{Exp}(-v/2)$$

$$\Pi_{BCH}(u) = u + [v, u] + \frac{1}{2}[v[v, u]]$$



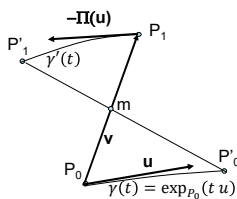
[ Lorenzi, Pennec: Efficient Parallel Transport of Deformations in Time Series of Images: from Schild's to pole Ladder, JMIV 50(1-2):5-17, 2013 ]

26/02/2025

48

## Parallel transport along geodesics

### Simpler scheme along geodesics: Pole Ladder



### Numerical accuracy of pole ladder

- Order 4 in general affine manifolds

$$\text{pole}(u) = \Pi(u) + \frac{1}{12} \nabla_v R(u, v)(5u - 2v) + \frac{1}{12} \nabla_u R(u, v)(v - 2u) + O(5)$$

- Error vanishes in symmetric spaces:

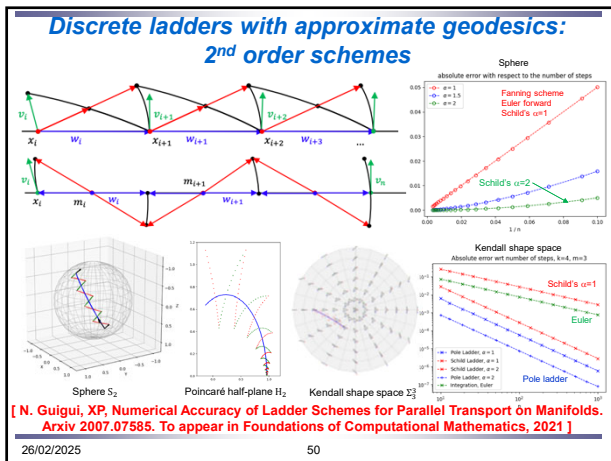
**Pole ladder is exact in 1 step!**

[ XP. Parallel Transport with Pole Ladder: a Third Order Scheme in Affine Connection Spaces which is Exact in Affine Symmetric Spaces. Arxiv 1805.11436 ]

26/02/2025

49






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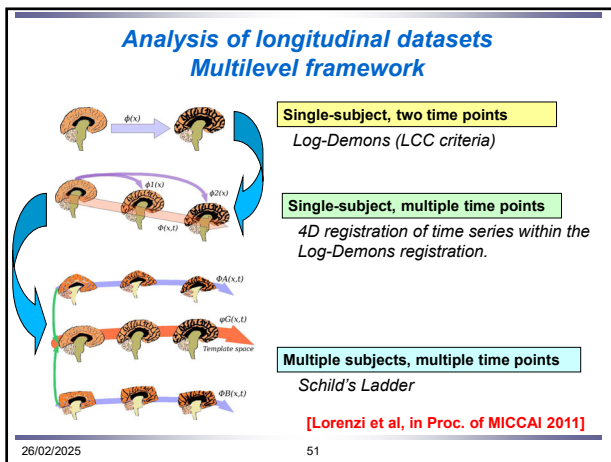
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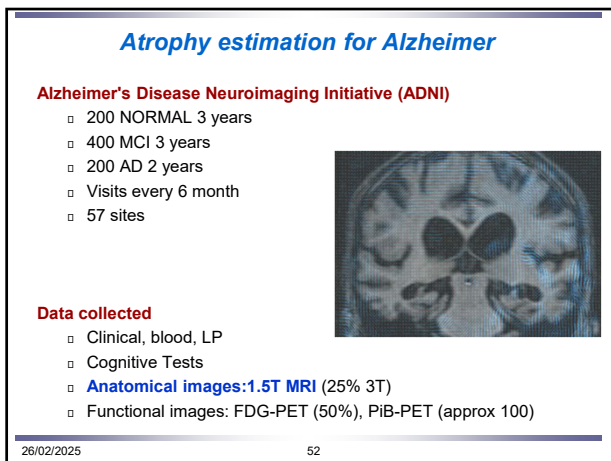
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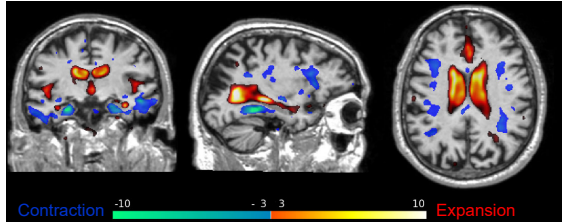
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### Modeling longitudinal atrophy in AD from images

#### One year structural changes for 70 Alzheimer's patients

- Median evolution model and significant atrophy (FdR corrected)



[Lorenzi et al, in Proc. of IPMI 2011]

26/02/2025

53

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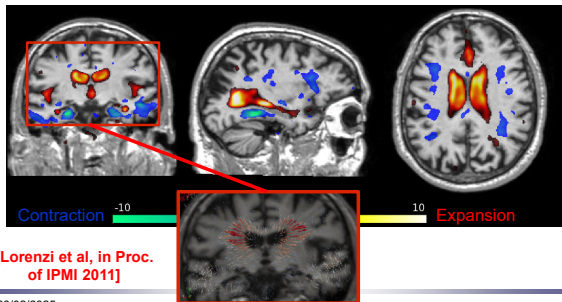
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26/02/2025

54

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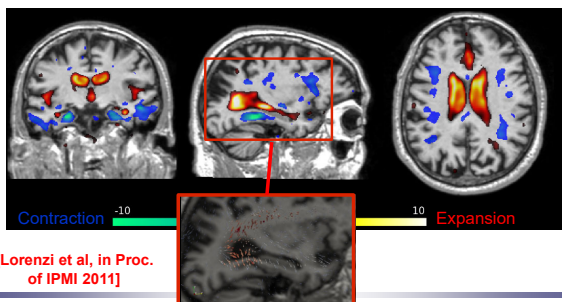
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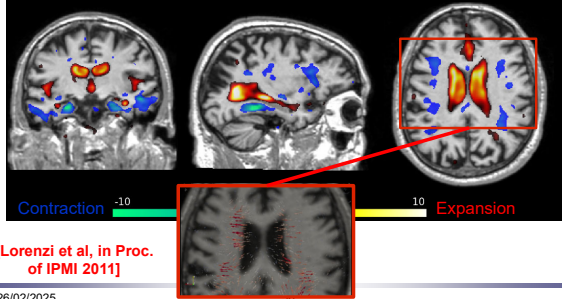
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### Modeling longitudinal atrophy in AD from images

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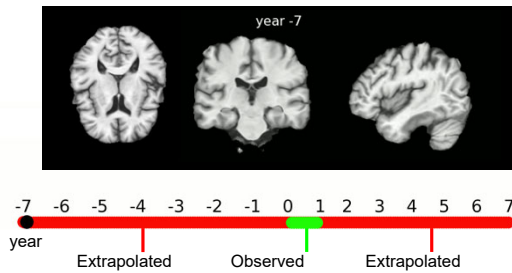
- Median evolution model and significant atrophy (FDR corrected)



26/02/2025

### Longitudinal model for AD

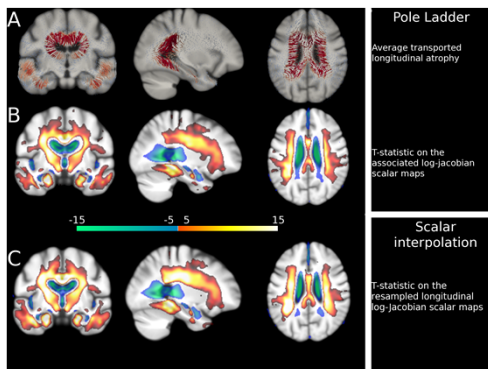
Estimated from 1 year changes – Extrapolation to 15 years  
70 AD subjects (ADNI data)



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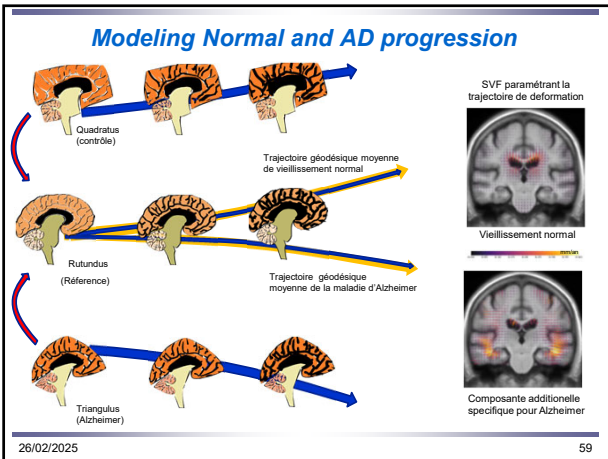
57

### Modeling longitudinal atrophy in AD from images



26/02/2025

58




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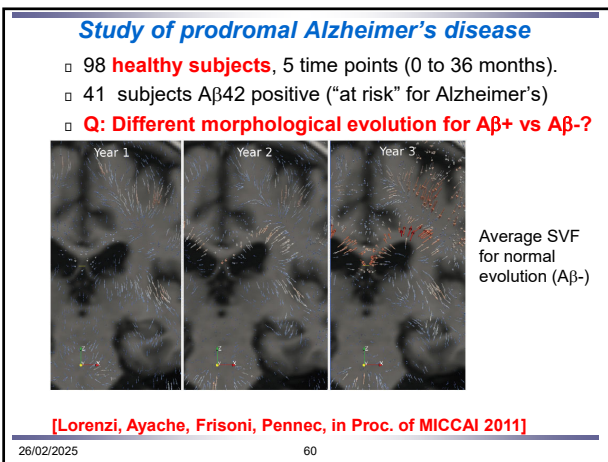
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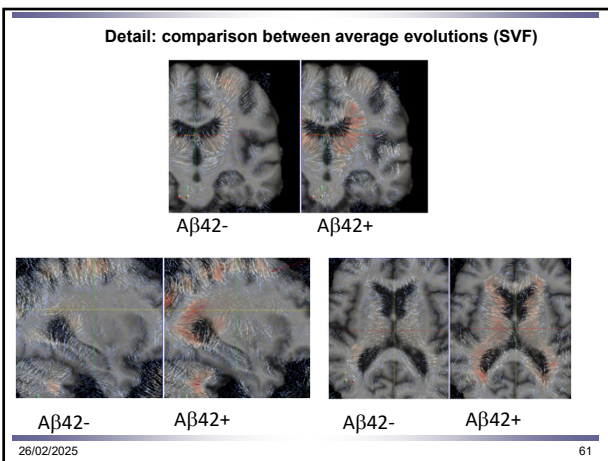
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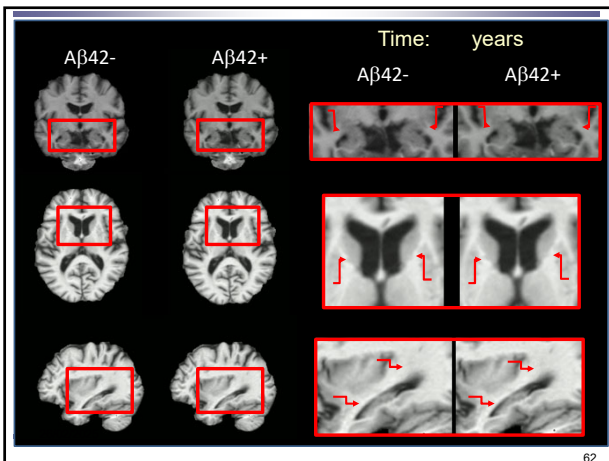
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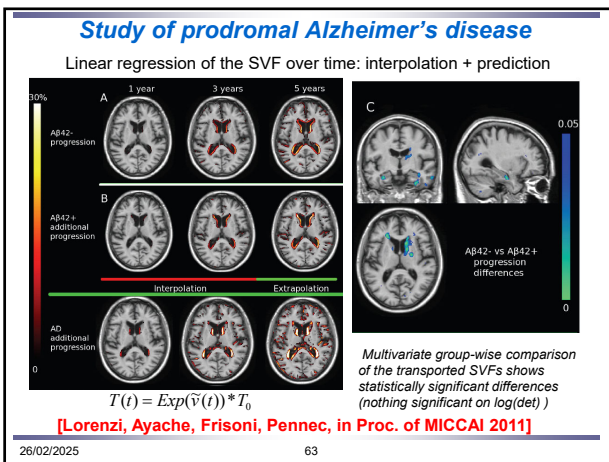
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**Statistical Computing on Manifolds for Computational Anatomy**

**Metric and Affine Geometric Settings for Lie Groups**

- Deformable image registration
- Riemannian frameworks on Lie groups
- Lie groups as affine connection spaces

**Modeling longitudinal deformations in AD**

- The SVF framework for diffeomorphisms
- Parallel transport of deformation trajectories

**Perspectives for statistics on deformation**

26/02/2025 6

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### The Stationnary Velocity Fields (SVF) framework for diffeomorphisms

- SVF framework for diffeomorphisms is algorithmically simple
- Compatible with "inverse-consistency"
- Vector statistics directly generalized to diffeomorphisms.

#### Registration algorithms using log-demons:

- Log-demons: Open-source ITK implementation (Vercauteren MICCAI 2008)  
<http://hdl.handle.net/10380/3060>  
[MICCAI Young Scientist Impact award 2013]
- Tensor (DTI) Log-demons (Sweet WBIR 2010):  
<https://gforge.inria.fr/projects/ttk>
- LCC log-demons for AD (Lorenzi, Neuroimage. 2013)  
<https://team.inria.fr/asclepios/software/lcclogdemons/>
- 3D myocardium strain / incompressible deformations (Mansi MICCAI'10)
- Hierarchical multiscale polyaffine log-demons (Seiler, Media 2012)  
<http://www.stanford.edu/~cseiler/software.html>  
[MICCAI 2011 Young Scientist award]

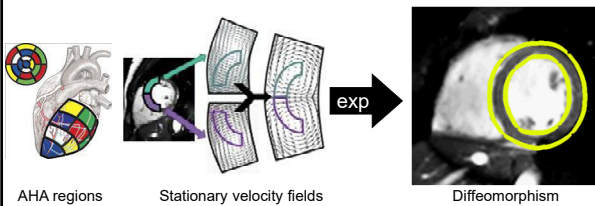
26/02/2025

65

### A powerful framework for statistics

#### Parametric diffeomorphisms [Arsigny et al., MICCAI 06, JMIV 09]

- One affine transformation per region (polyaffines transformations)
  - Cardiac motion tracking for **each subject** [McLeod, Miccai 2013]
- Log demons projected but with 204 parameters instead of a few millions



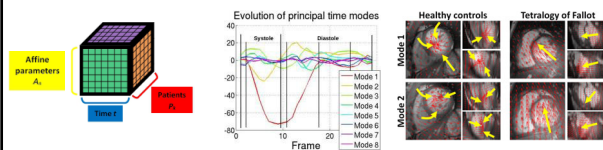
26/02/2025

66

### A powerful framework for statistics

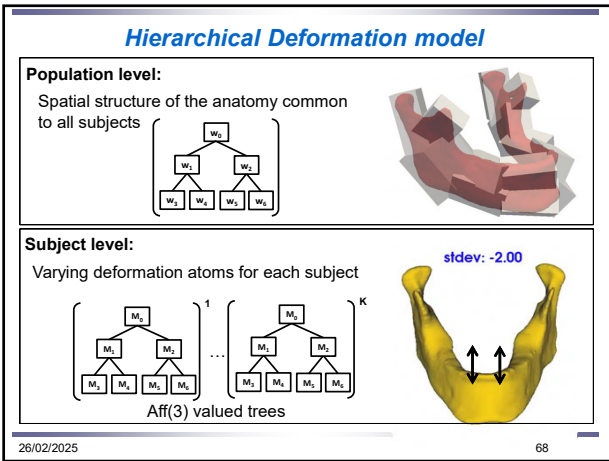
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- One affine transformation per region (polyaffines transformations)
  - Cardiac motion tracking for **each subject** [McLeod, Miccai 2013]
- Log demons projected but with 204 parameters instead of a few millions
- Group analysis** using tensor reduction : reduced model  
8 temporal modes x 3 spatial modes = 24 parameters (instead of 204)



26/02/2025

67




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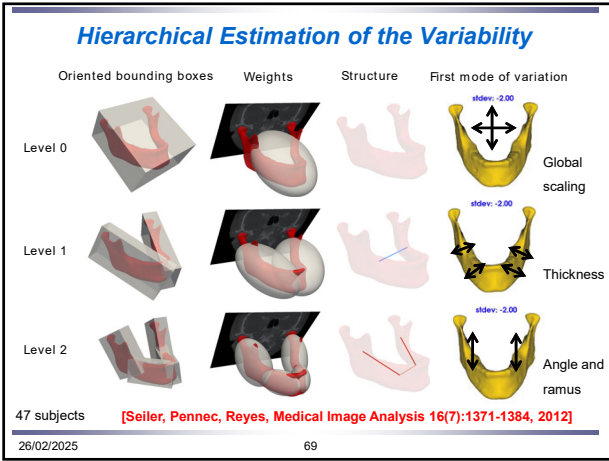
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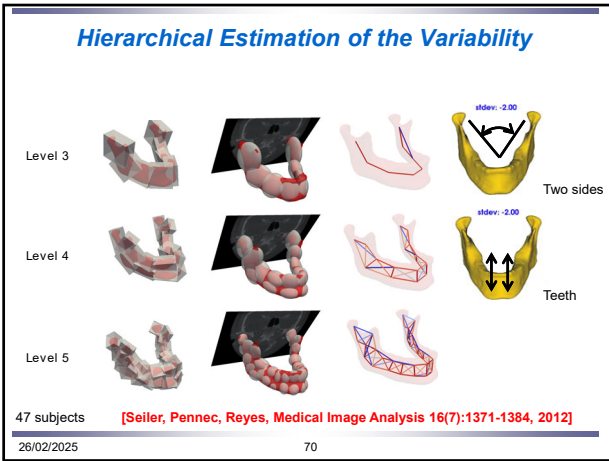
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